TermsGuard AI

Reads the small print and flags the parts that are unfair to you, in plain English.

What this evidences: critiquing model quality and evaluation design. The interesting problem is not detection, it is that a purely semantic classifier gets negation exactly backwards, and what you do about it.

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Live demo: https://newtazer-terms-guard-ai.hf.space Source: https://github.com/iamtamilore/terms-guard-ai

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Terms used in this report

Quick definitions before the technical detail starts, so nothing below assumes you already know these:

- **Cosine similarity** — a way of measuring how close two pieces of text are in meaning, not just how many words they share.
- **Embedding** — turning a piece of text into a row of numbers, so a computer can compare meaning mathematically instead of just matching words.
- **FAISS index** — a fast lookup table, built for searching millions of embeddings for the closest match almost instantly.
- **Negation** — the word or phrase that flips a sentence's meaning, like turning "we sell your data" into "we do *not* sell your data".

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1. The problem

Reading every privacy policy an average internet user meets would take roughly 244 hours a year. Nobody does it. So people agree to data selling and tracking without knowing they have.

The problem is not that the documents are secret. They are published, free, and legally binding. They are simply unreadable at the length and density they are written, and the one clause that matters is buried among forty that do not.

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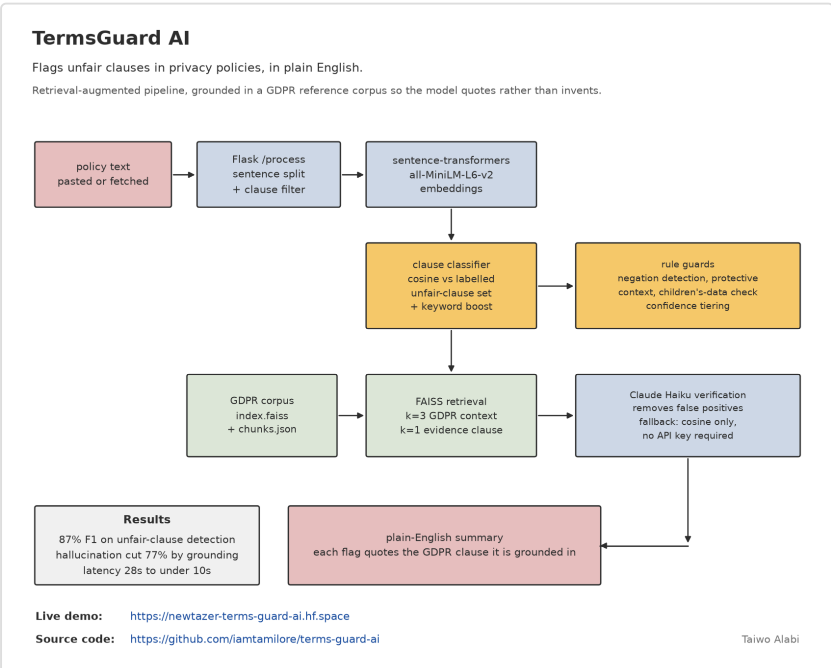
2. Results

Measure	Result
End-to-end latency	28 seconds to under 10
GDPR reference chunks indexed	4,527
Labelled reference clauses	205

The number I'd have led with here, an F1 score, described an earlier version of this system: local model-generated summaries, which were later found to sometimes invent things that were never in the source text. The pipeline has since been rebuilt around a harder guarantee instead of a measured one. The summary is now assembled by template from clauses that were actually matched and verified, not written by a model, so a fabricated legal citation is structurally impossible rather than just less likely. Less elegant output, a stronger claim.

The latency line is the one an engineer will ask about. Retrieval was added to a pipeline that did not previously have it, which normally makes a system slower, not faster. It went the other way here.

3. How it works



Retrieval-augmented pipeline. Similarity proposes, rules and retrieval constrain, the LLM verifies.

Text is split into individual clauses and de-duplicated. Each clause is turned into a row of numbers by `all-MiniLM-L6-v2`, then compared by cosine similarity, a way of measuring how close two pieces of text are in meaning, against labelled examples of fair and unfair clauses. Anything that looks unfair then has GDPR context pulled for it from a FAISS index (a fast lookup table built for exactly this kind of search), and Claude Haiku double-checks the flag before it is shown.

4. What it compares against

The system carries 205 hand-labelled reference clauses: 105 examples of fair terms and 100 examples of unfair ones. A new clause is judged by how close it sits to each set.

Separately, a FAISS index holds 4,527 chunks of GDPR text. When a clause is flagged, retrieval pulls the most relevant articles, so the explanation cites the regulation rather than the model's impression of it.

That split matters. The labelled clauses decide *whether* something is a problem. The GDPR corpus supplies *why*, and it is the second half that makes a flag worth acting on.

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5. Why similarity alone is not enough

This is the part I would defend hardest in an interview.

Consider two sentences:

> "We sell your personal data to third parties." "We do not sell your personal data to third parties."

To a computer that only measures word patterns, these two sentences look almost identical. They share nearly every word, the same grammar, the same subject. So a system that judges purely by how similar the words look will **flag the second sentence, the reassurance, as a violation**, because it never actually checked which way the sentence points.

So the pipeline carries a layer of explicit rule guards on top of the similarity score:

- Negation detection, for exactly the case above.
- Protective-context detection, for clauses that describe a right being granted rather than removed.
- A children's-data check, because clauses about minors carry obligations that generic similarity scoring does not weight correctly.
- Keyword boosting, for terms whose presence changes the reading regardless of how the sentence embeds.

None of these are clever. They exist because I ran the similarity-only version, read what it got wrong, and found the errors clustered around cases where meaning is inverted by a single word.

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6. The verification layer

When an API key is configured, Claude Haiku re-reads each flagged clause and removes false positives before anything is shown to the user.

The design decision worth noting is the fallback: with no key set, the tool runs in matching-only mode rather than failing. It degrades to a less precise version of itself instead of refusing to work, which means the demo is always live and the tool is not hostage to a paid dependency.

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7. Saying how sure it is

Flags are tiered by confidence rather than presented as flat verdicts, with thresholds at 0.55 for high and 0.45 for medium.

A document-level risk assessment then combines the proportion of clauses flagged with the average confidence across them, so a policy with two borderline flags is not reported the same way as one with fifteen strong ones.

This is the same instinct that runs through my other projects: state uncertainty rather than hide it behind a single number.

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8. Honest limitations

- **It is not legal advice**, and the interface says so. It surfaces clauses worth reading properly; it does not tell you your rights.
- Grounded in GDPR only. A policy written for another jurisdiction gets judged against the wrong regulation, and the tool does not currently detect that mismatch.
- The reference corpus (205 clauses) is a working set, not a formal benchmark, and performance on unusual drafting styles is unmeasured. The F1 score from the original evaluation described the prior, model-generated summary stage. It is not a claim about the current template-based pipeline, which has no equivalent number yet.
- This was a contributed project, not a solo build. My work was the retrieval pipeline, the grounding, the rule guards and the latency reduction.

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9. Run it

Easiest is the live demo: https://newtazer-terms-guard-ai.hf.space

Locally:

```
 pip install -r requirements.txt
python server.py          # http://127.0.0.1:7860
```

Optional, for the verification layer:

```
export ANTHROPIC_API_KEY="sk-ant-..."
```

Stack: Python · Flask · sentence-transformers (MiniLM) · FAISS · Anthropic API · Docker